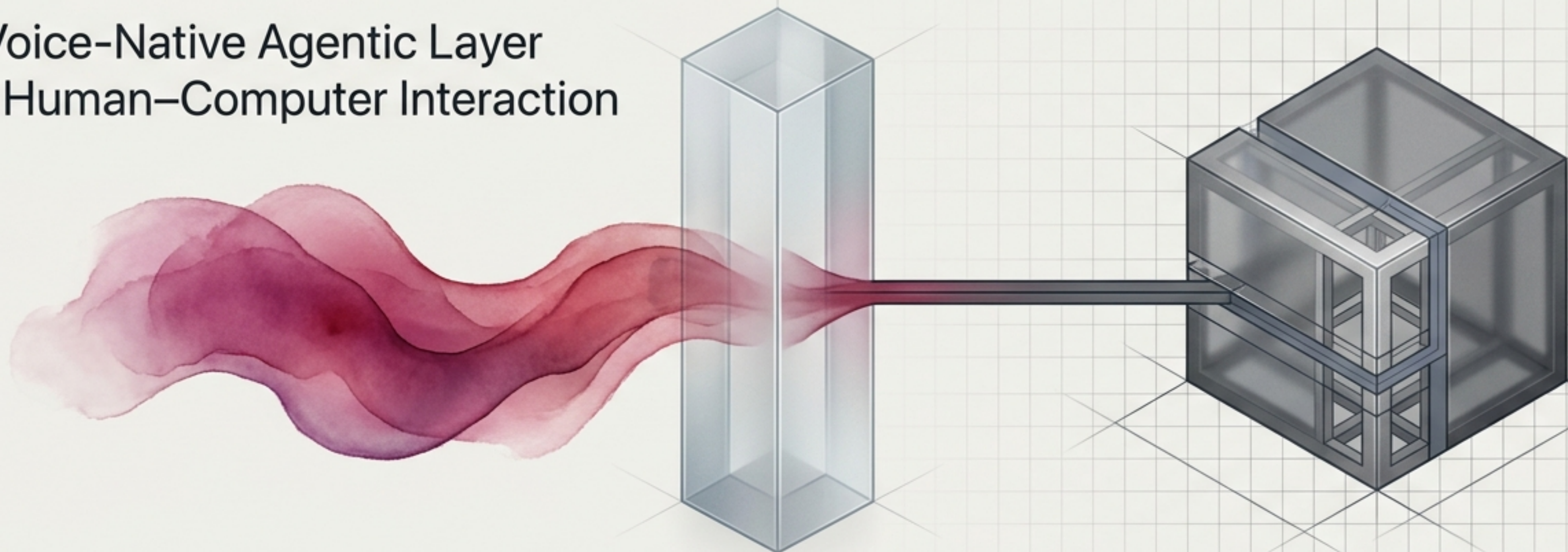


IntentOS

A Voice-Native Agentic Layer
for Human-Computer Interaction



From human intent to executable action.

Built for Qualcomm Snapdragon AI Lab.

The Interface Tax



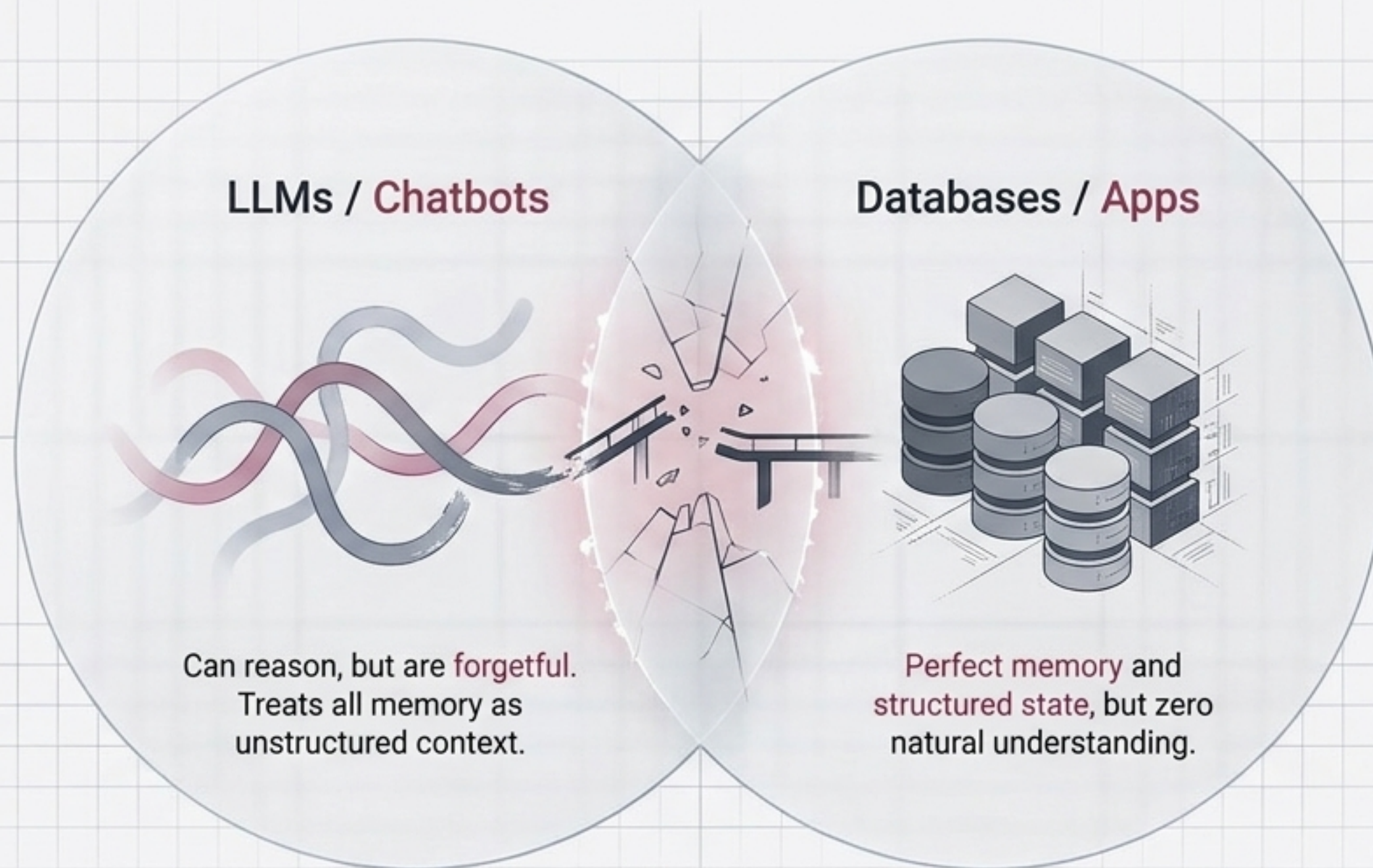
Today, users must act as the routing layer between their own ideas and their computer's structured data. To perform a task, users must remember which application to open, which screen to navigate, and how the information is structured.

Direct Execution



Humans think in goals. Software makes them think in interfaces.

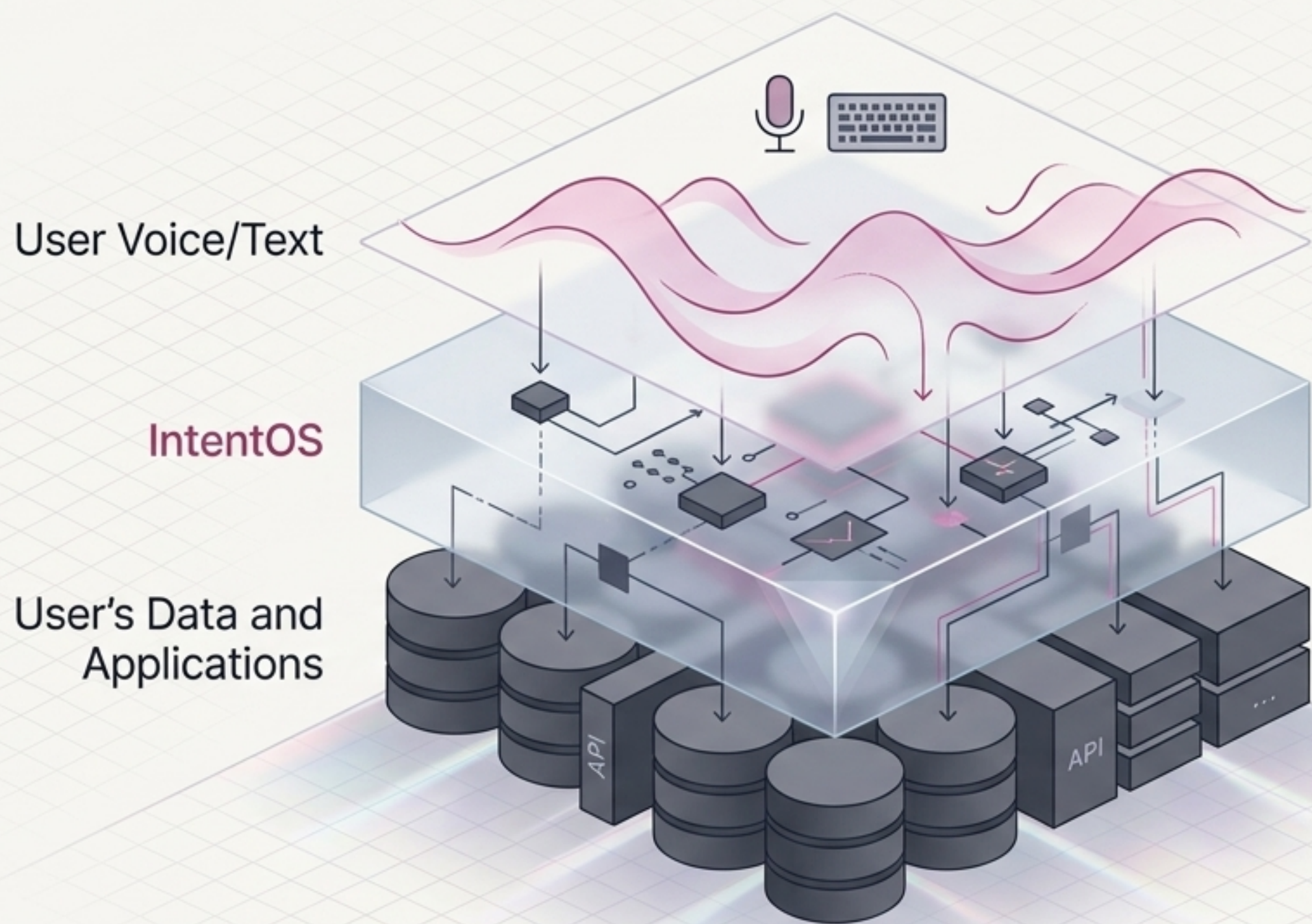
AI assistants are intelligent but forgetful.
Computers remember but don't understand.



We cannot bridge this gap by blindly dumping a user's entire digital history into an LLM context window.
It is highly inefficient, introduces severe latency, and creates massive privacy risks.

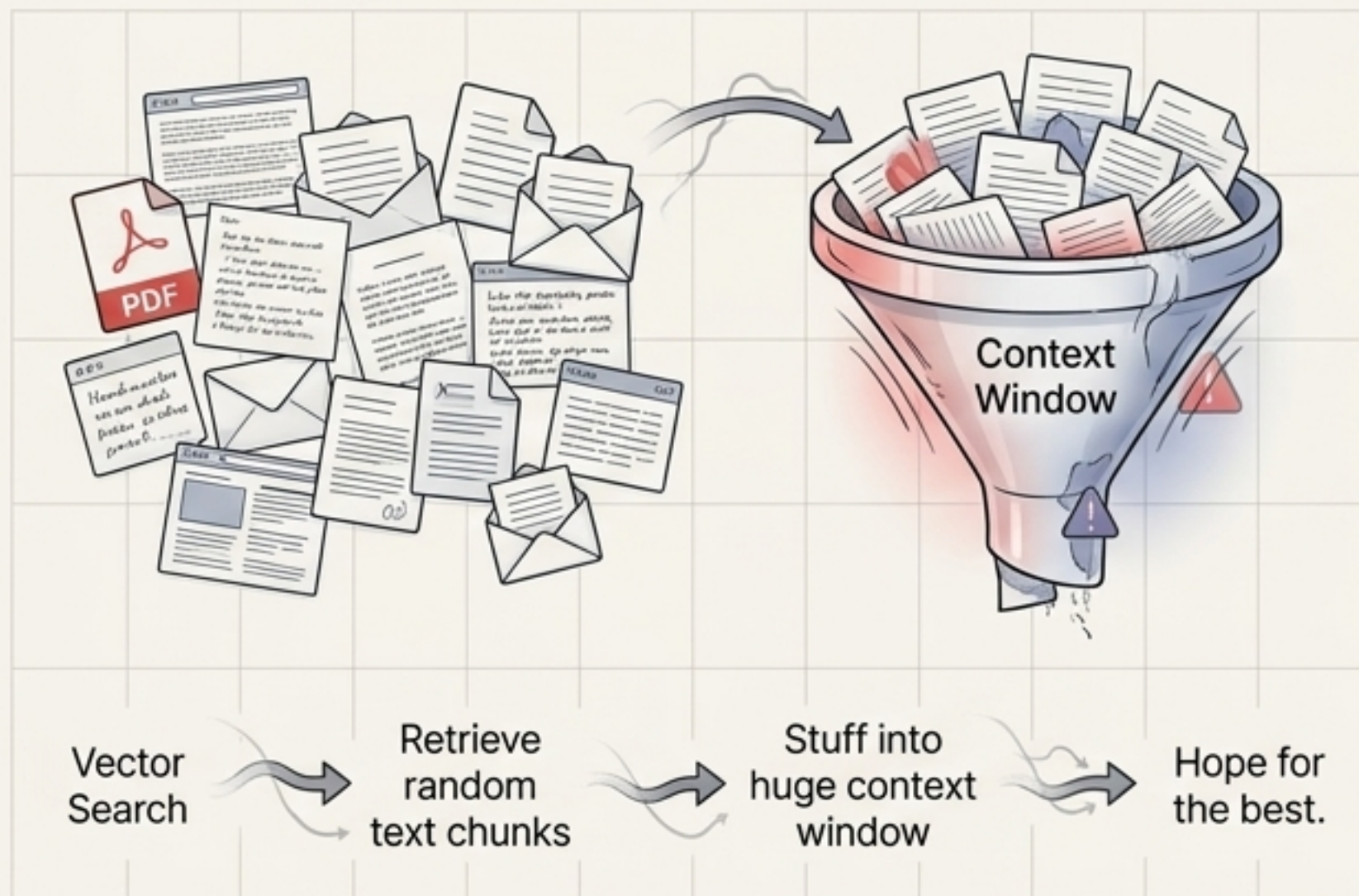
Introducing **IntentOS**: A local, voice-native agentic runtime layer.

IntentOS does not just generate text. It translates natural language into a **structured intent**, queries its own local memory, reasons over the retrieved context, executes real actions, and verifies the resulting state.

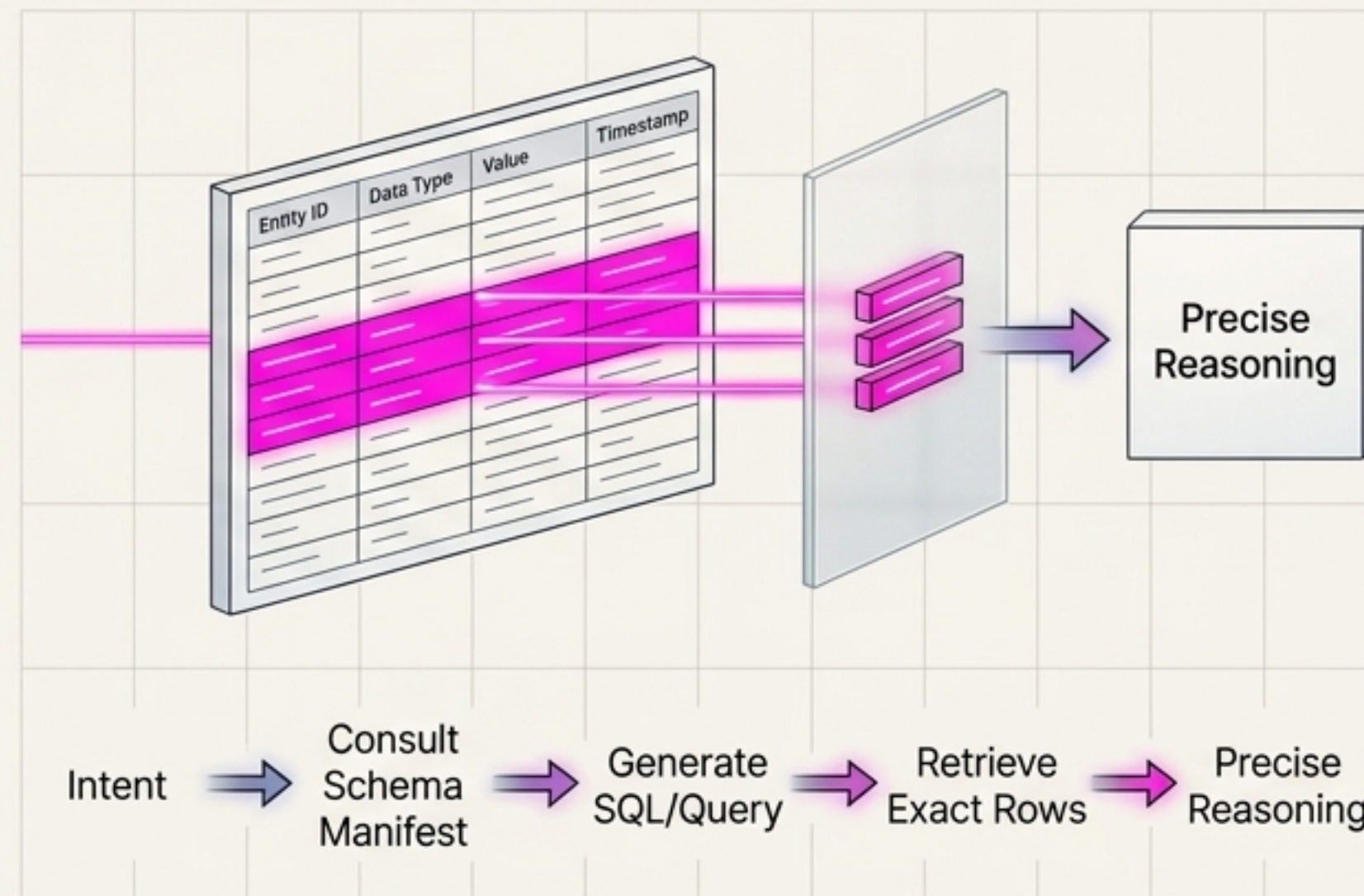


The AI doesn't search memory blindly. It understands the structure of its own memory.

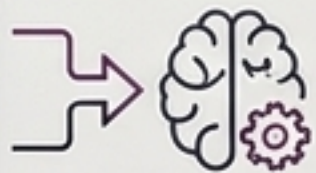
Traditional RAG: Blind Search



IntentOS: Structured Retrieval

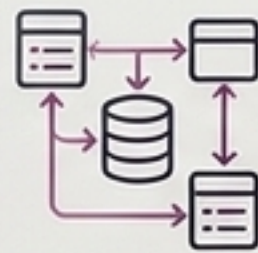


SQL as Context Compression. By using a schema manifest to query structured local databases (SQLite), IntentOS extracts only the required state. Don't make the local AI bigger—make the context smarter.



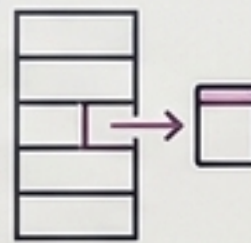
1. Intent-First Interaction

Users express goals; the system handles the software routing and tool selection.



2. Schema-Aware Memory

The AI inherently understands the relationships, tables, and rules of its underlying data.



3. Memory-Efficient Retrieval

Instead of loading entire histories, it executes narrow, highly specific state queries.



4. Action + Verification Loop

The agent does not stop at planning; it executes operations and actively verifies the state mutation.



5. Local-First Agent

Sensitive context and execution remain strictly on-device, ensuring absolute privacy.

Step 1: Understand

Voice input passes through the Intent Parser.

Step 2: Retrieve

The agent consults the Schema Manifest and queries the Local Memory Engine (SQLite).

Step 3: Reason

The local model plans the necessary action based on the retrieved state.

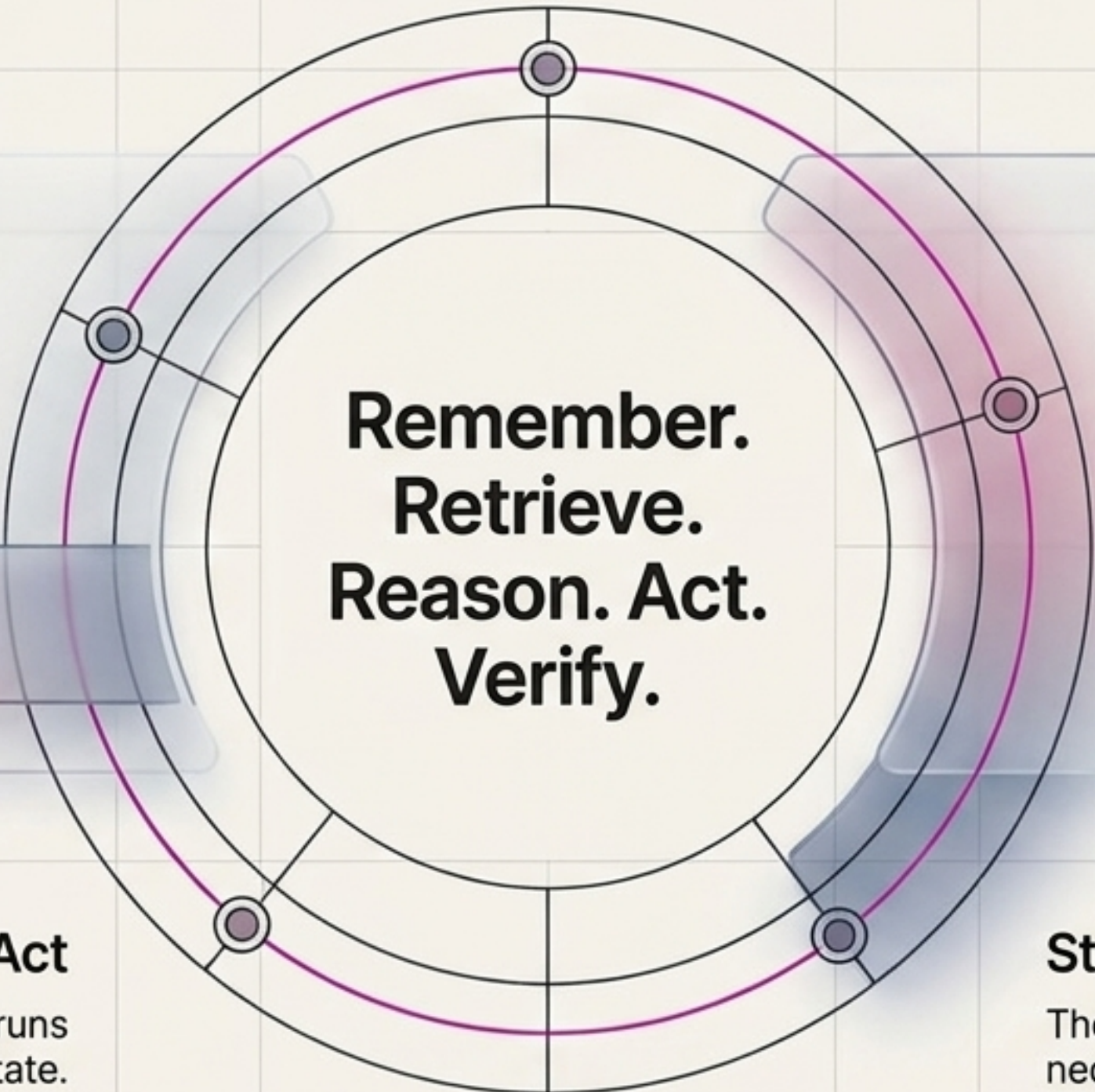
Step 4: Act

The Action Executor runs local tools to mutate state.

Step 5: Verify

The system checks the database to confirm state mutation before responding.

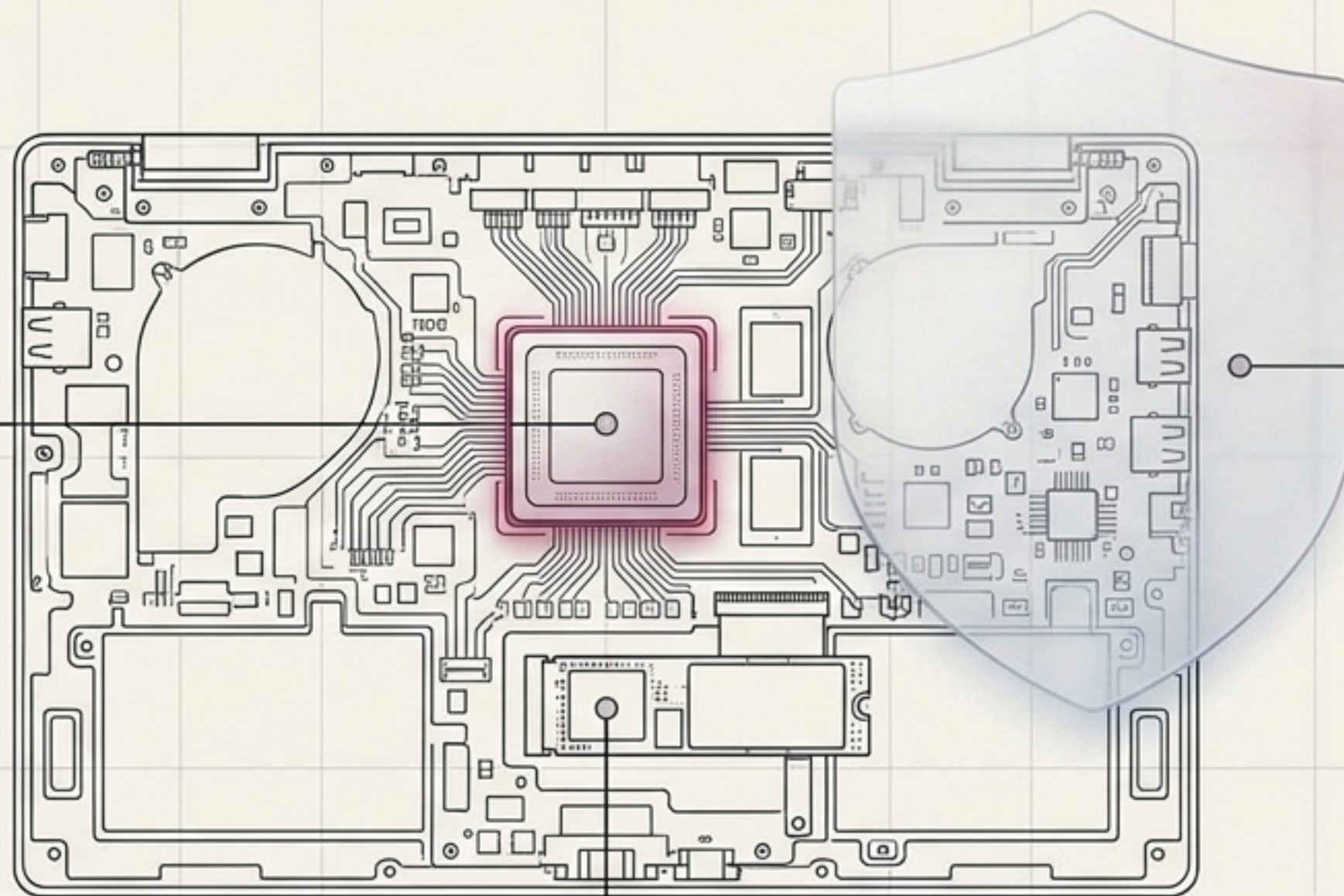
**Remember.
Retrieve.
Reason. Act.
Verify.**



Your intent. Your memory. Your actions. On your device.

Agentic Latency

Continuous intent-execution loops require zero-network-latency processing—ideal for NPU-accelerated workloads.



Privacy

Highly sensitive context (finances, routines, tasks) never leaves the device's local SQLite storage.

Context Efficiency

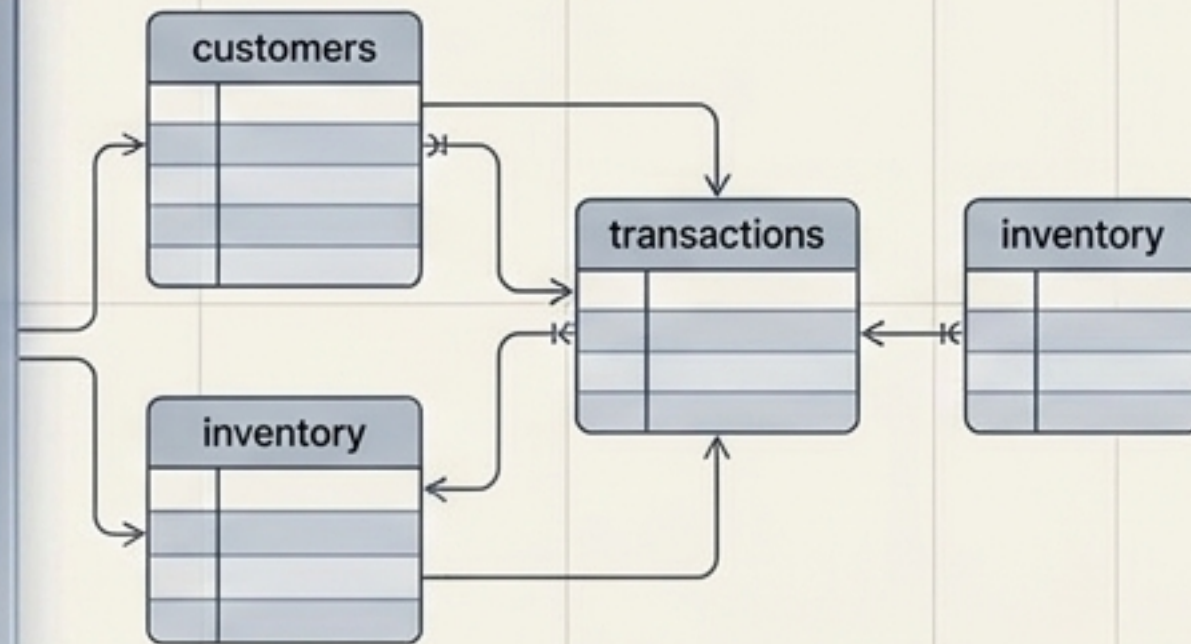
Schema-guided retrieval requires much smaller context windows, making it perfectly suited for highly optimized Small Language Models (SLMs) running locally.

IntentOS Platform
(General Runtime)

Validation Domain:
VyapaarSaathi
(Business Management)

We didn't just build a theory. We built a fully operational validation environment.

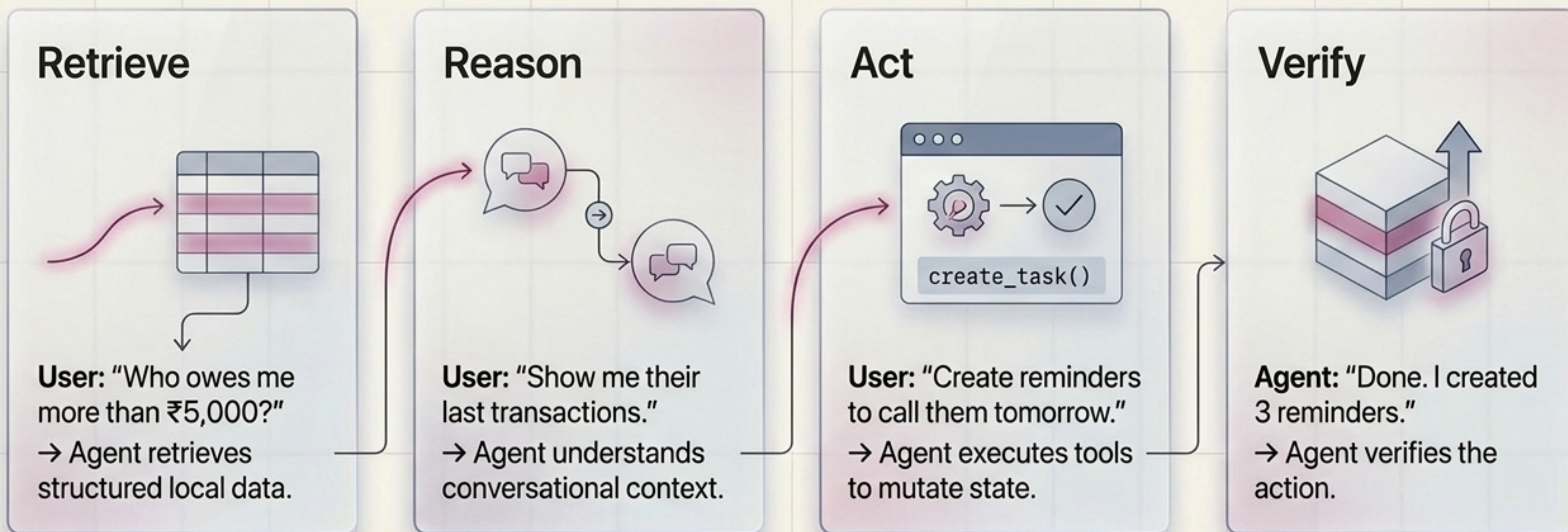
- SQLite persistence mapping complex business states.
- 18 distinct custom function tools for real state mutation.
- Transactional error handling and async backend integration.



```
// API Config: LiveKit Integration
const fivapps = gemini("Vesatsontosid");

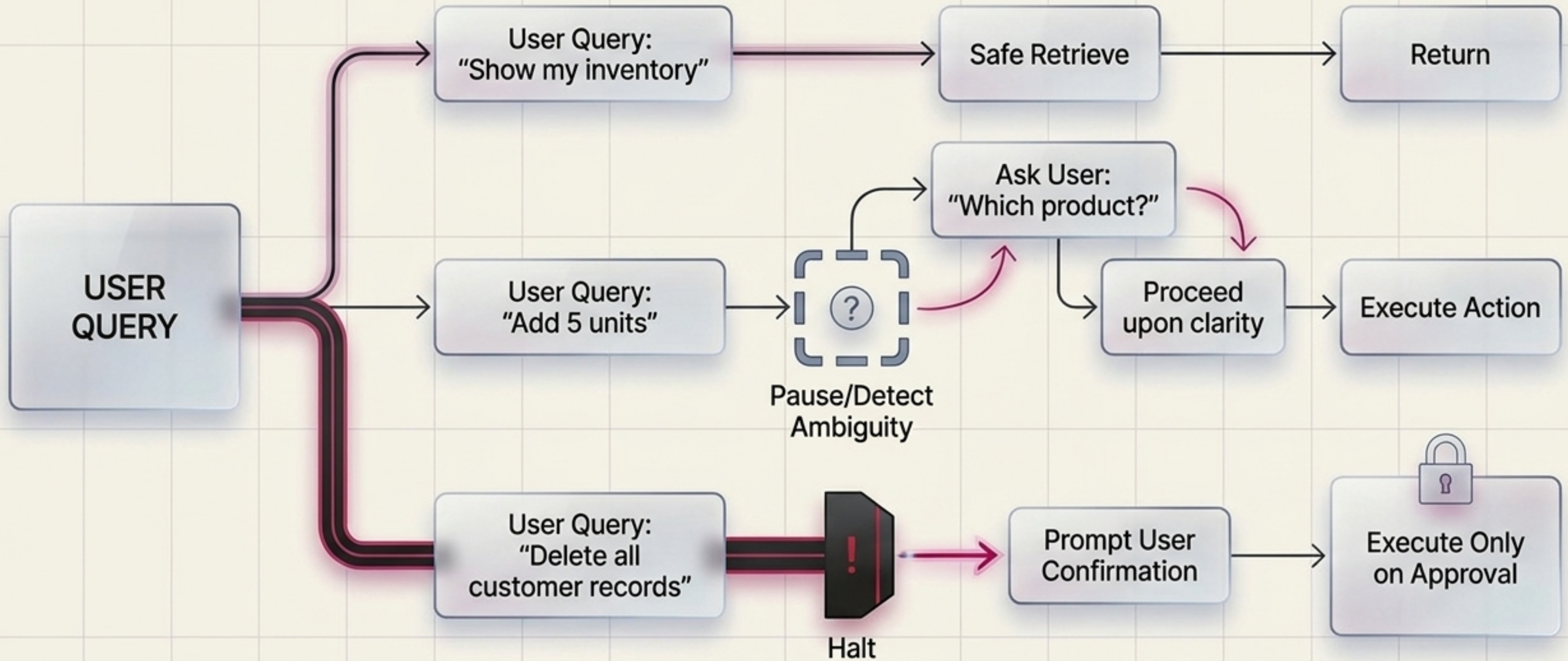
// Gemini Realtime Connection
name: "Rol",
parameter: "1$apWl",
apioneeroneoks:81,
parameterAame: input,
parameterType: "States",
configurator: "6on+ivdernare"
};

async function validateData(input) {
  ...
  return input;
}
```

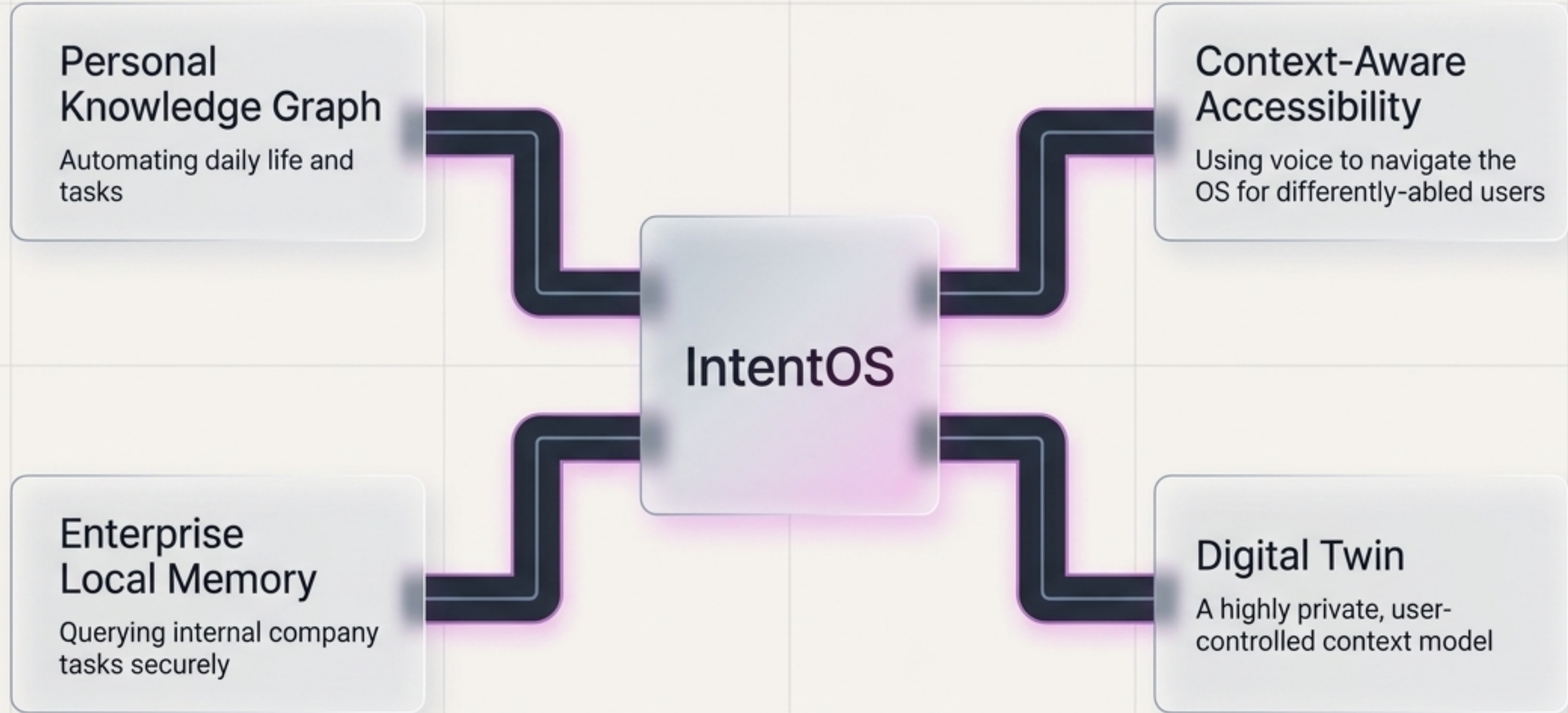


Demonstrating multi-step execution, persistent context, and verified action in a single, continuous flow.

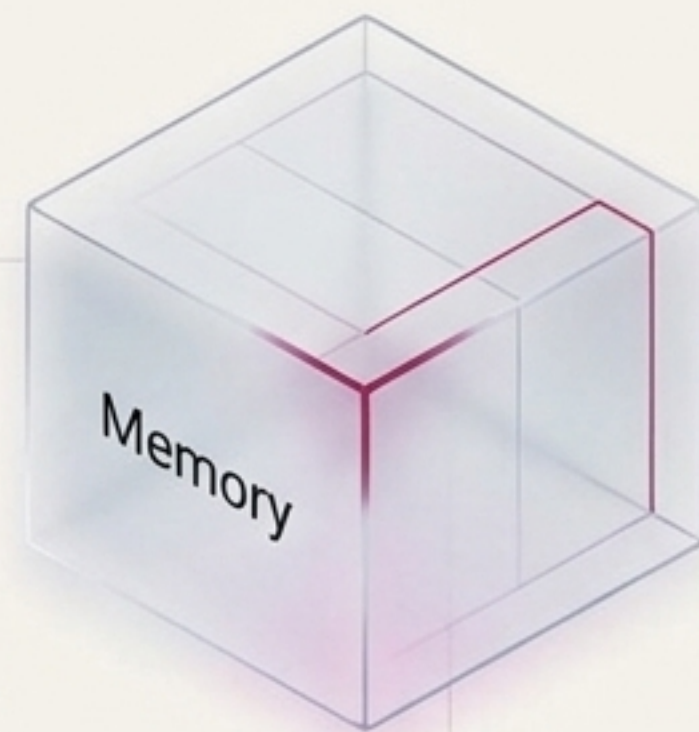
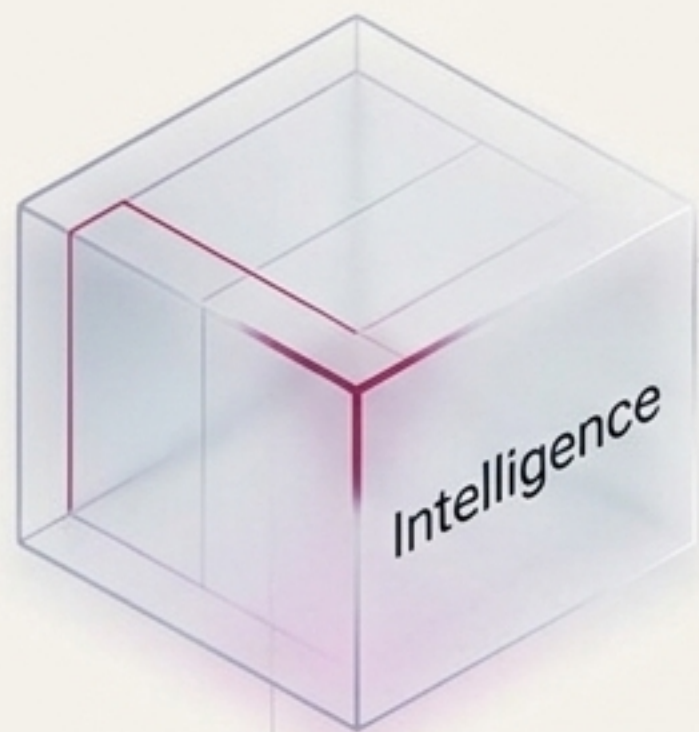
Act confidently when intent is clear; ask when ambiguity affects action.



Moving from a demonstration application to a contextual operating system for the AI PC era.

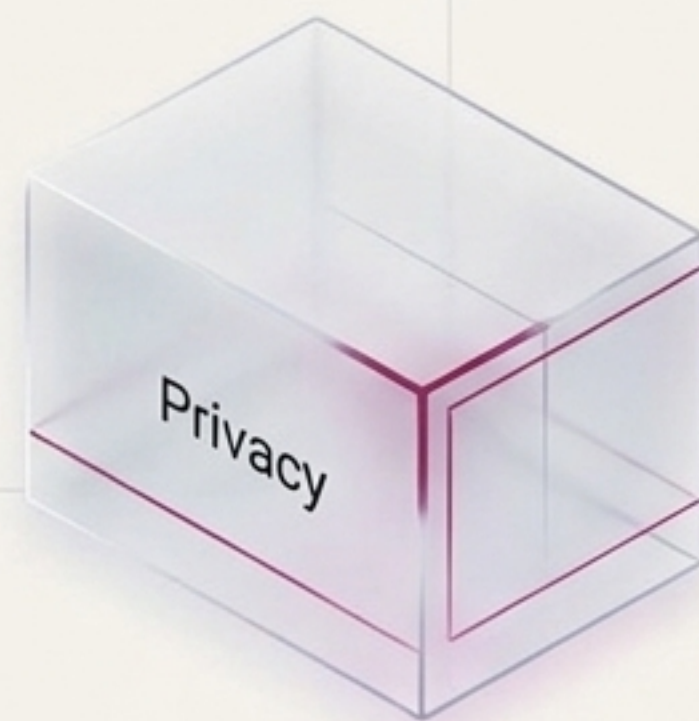
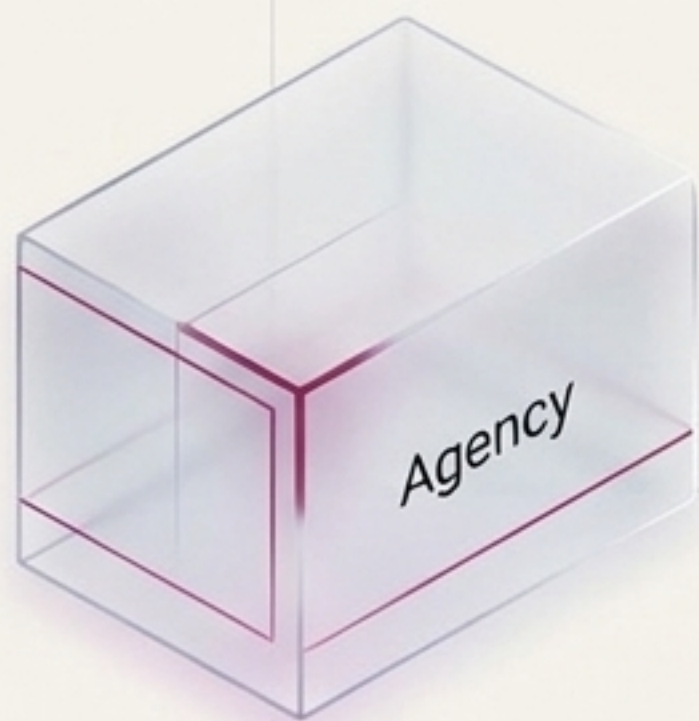


Swapping the domain requires only swapping the schema and tool registry.



A computer you can instruct instead of operate.

IntentOS gives AI a local structured memory it can query. It understands intent, retrieves only what it needs, executes real actions, and remembers the result—all locally on Snapdragon.



From human intent to executable action.